

LABORATORY REPORT

On

20MCA243 – MOBILE APPLICATION DEVELOPMENT LAB

THIRD SEMESTER

DEPARTMENT OF COMPUTER APPLICATIONS



NEHRU COLLEGE OF ENGINEERING AND RESEARCH CENTRE

AN AUTONOMOUS INSTITUTION AFFILIATED TO APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY, KERALA

(NAAC Re-Accredited with “A” Grade)

An ISO Certified Institution,

Pampady, Thiruvilwamala, Thrissur-680588

December 2024



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CERTIFICATE

Certified that this is the bonafide Laboratory report of the subject
20MCA243 Mobile Application Development Lab in Third Semester KTU-
MCA Practical done by.....

Register No:.....

during the academic year 2024-25

Lab in-charge

Head of the Department

Internal Examiner

External Examiner

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Program : 1

Android Application that uses GUI components, Font and Colors

Aim:

To develop a Simple Android Application that uses GUI components, Font and Colors.

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="30dp"
        android:gravity="center"
        android:text="TEAM MCA 2024"
        android:textSize="25sp"
        android:textStyle="bold"/>

    <Button
        android:id="@+id/button1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:gravity="center"
        android:text="Change font size"
        android:textSize="25sp"/>
    <Button
        android:id="@+id/button2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:gravity="center"
        android:text="Change color"
        android:textSize="25sp"/>
</LinearLayout>
```

Code for MainActivity.java:

```
import android.graphics.Color;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

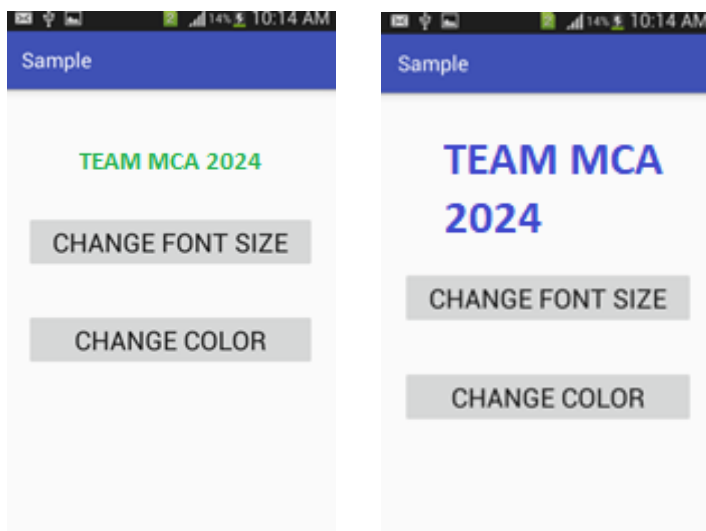
public class MainActivity extends AppCompatActivity
{
    int ch=1;
    float font=30;
    @Override
    Protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        final TextView t= (TextView) findViewById(R.id.textView);
        Button b1= (Button) findViewById(R.id.button1);
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
```

```

        Public void onClick(View v) {
            t.setTextSize(font);
            font = font + 5;
            if(font == 50)
                font = 30;
        }
    });
    Button b2= (Button) findViewById(R.id.button2);
    b2.setOnClickListener(new View.OnClickListener() {
        @Override
        Public void onClick(View v) {
            switch(ch) {
                case1:
                    t.setTextColor(Color.RED);
                    break;
                case2:
                    t.setTextColor(Color.GREEN);
                    break;
                case3:
                    t.setTextColor(Color.BLUE);
                    break;
                case4:
                    t.setTextColor(Color.CYAN);
                    break;
                case5:
                    t.setTextColor(Color.YELLOW);
                    break;
                case6:
                    t.setTextColor(Color.MAGENTA);
                    break;
            }
            ch++;
            if(ch == 7)
                ch = 1;
        }
    });
}

```

Output :



Program : 2

Simple Android Application for Native Calculator

Aim:

To develop a Simple Android Application for Native Calculator.

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="20dp">

    <LinearLayout
        android:id="@+id/linearLayout1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp">

        <EditText
            android:id="@+id/editText1"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:inputType="numberDecimal"
            android:textSize="20sp"/>

        <EditText
            android:id="@+id/editText2"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:inputType="numberDecimal"
            android:textSize="20sp"/>

    </LinearLayout>

    <LinearLayout
        android:id="@+id/linearLayout2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp">

        <Button
            android:id="@+id/Add"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="+"
            android:textSize="30sp"/>

        <Button
            android:id="@+id/Sub"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="-"
            android:textSize="30sp"/>

    </LinearLayout>

</LinearLayout>
```

```

        <Button
            android:id="@+id/Mul"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="*"
            android:textSize="30sp"/>

        <Button
            android:id="@+id/Div"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="/"
            android:textSize="30sp"/>

    </LinearLayout>

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:text="Answer is"
        android:textSize="30sp"
        android:gravity="center"/>

</LinearLayout>

```

Code for MainActivity.java:

```

package com.example.mca.calc;
import android.os.Bundle;
import android.support.v7.app.AppCompatActivity;
import android.text.TextUtils;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity implements OnClickListener
{
    //Defining the Views
    EditText Num1;
    EditText Num2;
    Button Add;
    Button Sub;
    Button Mul;
    Button Div;
    TextView Result;

    @Override
    public void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //Referring the Views
        Num1 = (EditText) findViewById(R.id.editText1);
        Num2 = (EditText) findViewById(R.id.editText2);
        Add = (Button) findViewById(R.id.Add);
        Sub = (Button) findViewById(R.id.Sub);
        Mul = (Button) findViewById(R.id.Mul);
        Div = (Button) findViewById(R.id.Div);
        Result = (TextView) findViewById(R.id.textView);
    }
}

```

```

        // set a listener
        Add.setOnClickListener(this);
        Sub.setOnClickListener(this);
        Mul.setOnClickListener(this);
        Div.setOnClickListener(this);
    }

    @Override
    public void onClick (View v)
    {

        float num1 = 0;
        float num2 = 0;
        float result = 0;
        String oper = "";

        // check if the fields are empty
        if(TextUtils.isEmpty(Num1.getText().toString()) ||
        TextUtils.isEmpty(Num2.getText().toString()))
            return;

        // read EditText and fill variables with numbers
        num1 = Float.parseFloat(Num1.getText().toString());
        num2 = Float.parseFloat(Num2.getText().toString());

        // defines the button that has been clicked and performs the
        corresponding operation
        // write operation into oper, we will use it later for output
        switch(v.getId())
        {
            case R.id.Add:
                oper = "+";
                result = num1 + num2;
                break;
            case R.id.Sub:
                oper = "-";
                result = num1 - num2;
                break;
            case R.id.Mul:
                oper = "*";
                result = num1 * num2;
                break;
            case R.id.Div:
                oper = "/";
                result = num1 / num2;
                break;
            default:
                break;
        }
        // form the output line
        Result.setText(num1 + " " + oper + " " + num2 + " = " + result);
    }
}

```


Output :

Calc

6040

+

-

*

/

60.0 * 40.0 = 2400.0

Calc

6040

+

-

*

/

60.0 - 40.0 = 20.0

Calc

2030

+

-

*

/

20.0 + 30.0 = 50.0

Calc

6040

+

-

*

/

60.0 / 40.0 = 1.5

Program : 3

Android Application to draw Basic Graphical Primitives

Aim:

To develop a Simple Android Application that draws basic Graphical Primitives on the screen.

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/imageView"/>
</RelativeLayout>
```

Code for MainActivity.java:

```
import android.app.Activity;
import android.graphics.Bitmap;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.graphics.drawable.BitmapDrawable;
import android.os.Bundle;
import android.widget.ImageView;

public class MainActivity extends Activity
{
    @Override
    public void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //Creating a Bitmap
        Bitmap bg = Bitmap.createBitmap(720, 1280, Bitmap.Config.ARGB_8888);

        //Setting the Bitmap as background for the ImageView
        ImageView i = (ImageView) findViewById(R.id.imageView);
        i.setBackgroundDrawable(new BitmapDrawable(bg));

        //Creating the Canvas Object
        Canvas canvas = new Canvas(bg);

        //Creating the Paint Object and set its color & TextSize
        Paint paint = new Paint();
        paint.setColor(Color.BLUE);
        paint.setTextSize(50);

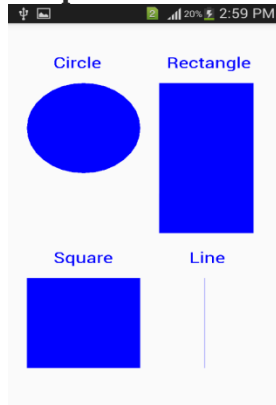
        //To draw a Rectangle
        canvas.drawText("Rectangle", 420, 150, paint);
        canvas.drawRect(400, 200, 650, 700, paint);

        //To draw a Circle
        canvas.drawText("Circle", 120, 150, paint);
        canvas.drawCircle(200, 350, 150, paint);
    }
}
```

```
//To draw a Square
canvas.drawText("Square", 120, 800, paint);
canvas.drawRect(50, 850, 350, 1150, paint);

//To draw a Line
canvas.drawText("Line", 480, 800, paint);
canvas.drawLine(520, 850, 520, 1150, paint);
}
```

Output :



Program : 4

Simple Android Application For Student Mark Entry that makes use of Database

Aim:

To develop a Simple Android Application that makes use of Database.

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="50dp"
        android:layout_y="20dp"
        android:text="Student Details"
        android:textSize="30sp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="110dp"
        android:text="Enter Rollno:"
        android:textSize="20sp"/>

    <EditText
        android:id="@+id/Rollno"
        android:layout_width="150dp"
        android:layout_height="wrap_content"
        android:layout_x="175dp"
        android:layout_y="100dp"
        android:inputType="number"
        android:textSize="20sp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="160dp"
        android:text="Enter Name:"
        android:textSize="20sp"/>

    <EditText
        android:id="@+id/Name"
        android:layout_width="150dp"
        android:layout_height="wrap_content"
        android:layout_x="175dp"
        android:layout_y="150dp"
        android:inputType="text"
        android:textSize="20sp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="210dp"
        android:text="Enter Marks:"
        android:textSize="20sp"/>

    <EditText
```

```

        android:id="@+id/Marks"
        android:layout_width="150dp"
        android:layout_height="wrap_content"
        android:layout_x="175dp"
        android:layout_y="200dp"
        android:inputType="number"
        android:textSize="20sp"/>

<Button
    android:id="@+id/Insert"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="300dp"
    android:text="Insert"
    android:textSize="30dp"/>

<Button
    android:id="@+id/Delete"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="300dp"
    android:text="Delete"
    android:textSize="30dp"/>

<Button
    android:id="@+id/Update"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="400dp"
    android:text="Update"
    android:textSize="30dp"/>

<Button
    android:id="@+id/View"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="400dp"
    android:text="View"
    android:textSize="30dp"/>

<Button
    android:id="@+id/ViewAll"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:layout_x="100dp"
    android:layout_y="500dp"
    android:text="View All"
    android:textSize="30dp"/>

</AbsoluteLayout>

```

Code for MainActivity.java:

```

import android.app.Activity;
import android.app.AlertDialog.Builder;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;

```

```

public class MainActivity extends Activity implements OnClickListener
{
    EditText Rollno,Name,Marks;
    Button Insert,Delete,Update,View,ViewAll;
    SQLiteDatabase db;
    /** Called when the activity is first created. */
    @Override
    public void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Rollno=(EditText) findViewById(R.id.Rollno);
        Name=(EditText) findViewById(R.id.Name);
        Marks=(EditText) findViewById(R.id.Marks);
        Insert=(Button) findViewById(R.id.Insert);
        Delete=(Button) findViewById(R.id.Delete);
        Update=(Button) findViewById(R.id.Update);
        View=(Button) findViewById(R.id.View);
        ViewAll=(Button) findViewById(R.id.ViewAll);

        Insert.setOnClickListener(this);
        Delete.setOnClickListener(this);
        Update.setOnClickListener(this);
        View.setOnClickListener(this);
        ViewAll.setOnClickListener(this);

        // Creating database and table
        db=openOrCreateDatabase("StudentDB", Context.MODE_PRIVATE, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS student(rollno VARCHAR,name
VARCHAR,marks VARCHAR);");
    }
    public void onClick(View view)
    {
        // Inserting a record to the Student table
        if(view==Insert)
        {
            // Checking for empty fields
            if(Rollno.getText().toString().trim().length()==0||
                Name.getText().toString().trim().length()==0||
                Marks.getText().toString().trim().length()==0)
            {
                showMessage("Error", "Please enter all values");
                return;
            }
            db.execSQL("INSERT INTO student
VALUES('"+Rollno.getText()+"','"+Name.getText()+"
            "','"+Marks.getText()+"');");
            showMessage("Success", "Record added");
            clearText();
        }
        // Deleting a record from the Student table
        if(view==Delete)
        {
            // Checking for empty roll number
            if(Rollno.getText().toString().trim().length()==0)
            {
                showMessage("Error", "Please enter Rollno");
                return;
            }
            Cursor c=db.rawQuery("SELECT * FROM student WHERE
rollno='"+Rollno.getText()+"'", null);
            if(c.moveToFirst())
            {
                db.execSQL("DELETE FROM student WHERE
rollno='"+Rollno.getText()+"'");
                showMessage("Success", "Record Deleted");
            }
            else
            {

```

```

        showMessage("Error", "Invalid Rollno");
    }
    clearText();
}
// Updating a record in the Student table
if(view==Update)
{
    // Checking for empty roll number
    if(Rollno.getText().toString().trim().length()==0)
    {
        showMessage("Error", "Please enter Rollno");
        return;
    }
    Cursor c=db.rawQuery("SELECT * FROM student WHERE
rollno='"+Rollno.getText()+"'", null);
    if(c.moveToFirst()) {
        db.execSQL("UPDATE student SET name='"+ Name.getText() +
"',marks='"+ Marks.getText() +
        "' WHERE rollno='"+Rollno.getText()+"'");
        showMessage("Success", "Record Modified");
    }
    else{
        showMessage("Error", "Invalid Rollno");
    }
    clearText();
}
// Display a record from the Student table
if(view==View)
{
    // Checking for empty roll number
    if(Rollno.getText().toString().trim().length()==0)
    {
        showMessage("Error", "Please enter Rollno");
        return;
    }
    Cursor c=db.rawQuery("SELECT * FROM student WHERE
rollno='"+Rollno.getText()+"'", null);
    if(c.moveToFirst())
    {
        Name.setText(c.getString(1));
        Marks.setText(c.getString(2));
    }
    else
    {
        showMessage("Error", "Invalid Rollno");
        clearText();
    }
}
// Displaying all the records
if(view==ViewAll)
{
    Cursor c=db.rawQuery("SELECT * FROM student", null);
    if(c.getCount()==0)
    {
        showMessage("Error", "No records found");
        return;
    }
    StringBuffer buffer=newStringBuffer();
    while(c.moveToNext())
    {
        buffer.append("Rollno: "+c.getString(0)+"\n");
        buffer.append("Name: "+c.getString(1)+"\n");
        buffer.append("Marks: "+c.getString(2)+"\n\n");
    }
    showMessage("Student Details", buffer.toString());
}
}
public void showMessage(String title,String message)
{
    Builder builder=newBuilder(this);
    builder.setCancelable(true);

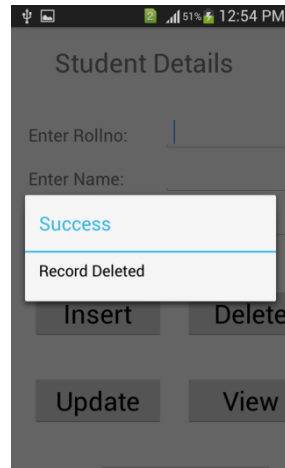
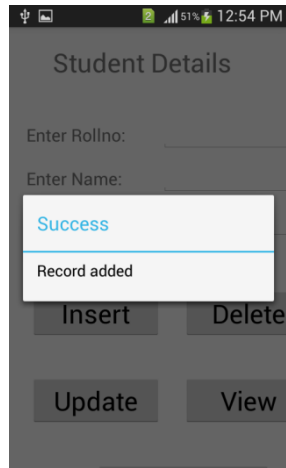
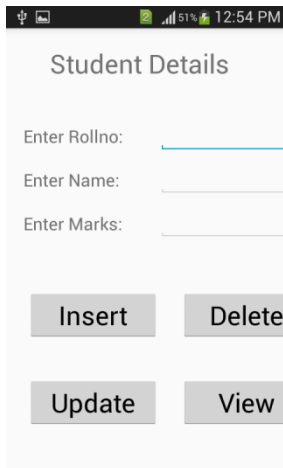
```

```

        builder.setTitle(title);
        builder.setMessage(message);
        builder.show();
    }
    public void clearText()
    {
        Rollno.setText("");
        Name.setText("");
        Marks.setText("");
        Rollno.requestFocus();
    }
}

```

Output :



Program : 5

Android Application that implements Multi threading

Aim:

To develop an Android Application that implements Multi threading.

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="250dp"
        android:layout_height="250dp"
        android:layout_margin="50dp"
        android:layout_gravity="center"/>

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_gravity="center"
        android:text="Load Image 1"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_gravity="center"
        android:text="Load image 2"/>

</LinearLayout>
```

Code for MainActivity.java:

```
import android.os.Bundle;
import android.support.v7.app.AppCompatActivity;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
public class MainActivity extends AppCompatActivity
{
    ImageView img;
    Button bt1, bt2;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        bt1 = (Button) findViewById(R.id.button);
        bt2 = (Button) findViewById(R.id.button2);
        img = (ImageView) findViewById(R.id.imageView);

        bt1.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
```

```

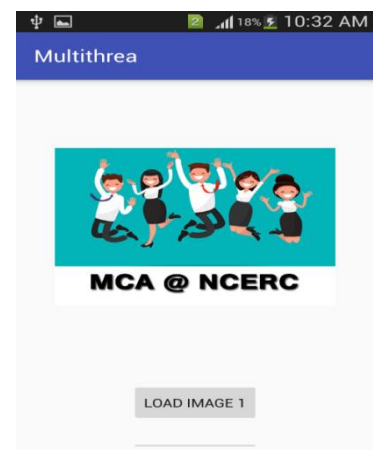
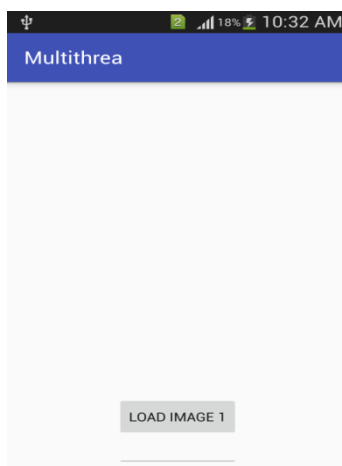
        new Thread(new Runnable()
        {
            @Override
            public void run()
            {
                img.post(new Runnable()
                {
                    @Override
                    public void run()
                    {
                        img.setImageResource(R.drawable.abc);
                    }
                });
            }
        }).start();
    }
});

bt2.setOnClickListener(new View.OnClickListener()
{
    @Override
    public void onClick(View v)
    {
        new Thread(new Runnable()
        {
            @Override
            public void run()
            {
                img.post(new Runnable()
                {
                    @Override
                    public void run()
                    {
                        img.setImageResource(R.drawable.cba);
                    }
                });
            }
        }).start();
    }
});
}
}
}

```

Note: Before Running the Application, Copy the Images and Paste it in “app -> res -> drawable” by pressing “right click mouse button on drawable” and selecting the “Paste” option.

Output :



Program : 6

Android Application that creates an alert upon receiving a message

Aim:

To develop an Android Application that creates an alert upon receiving a message.

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Message"
        android:textSize="30sp"/>

    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:singleLine="true"
        android:textSize="30sp"/>

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="30dp"
        android:layout_gravity="center"
        android:text="Notify"
        android:textSize="30sp"/>

</LinearLayout>
```

Code for MainActivity.java:

```
import android.app.Notification;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Intent;
import android.os.Bundle;
import android.support.v7.app.AppCompatActivity;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity
{
    Button notify;
    EditText e;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

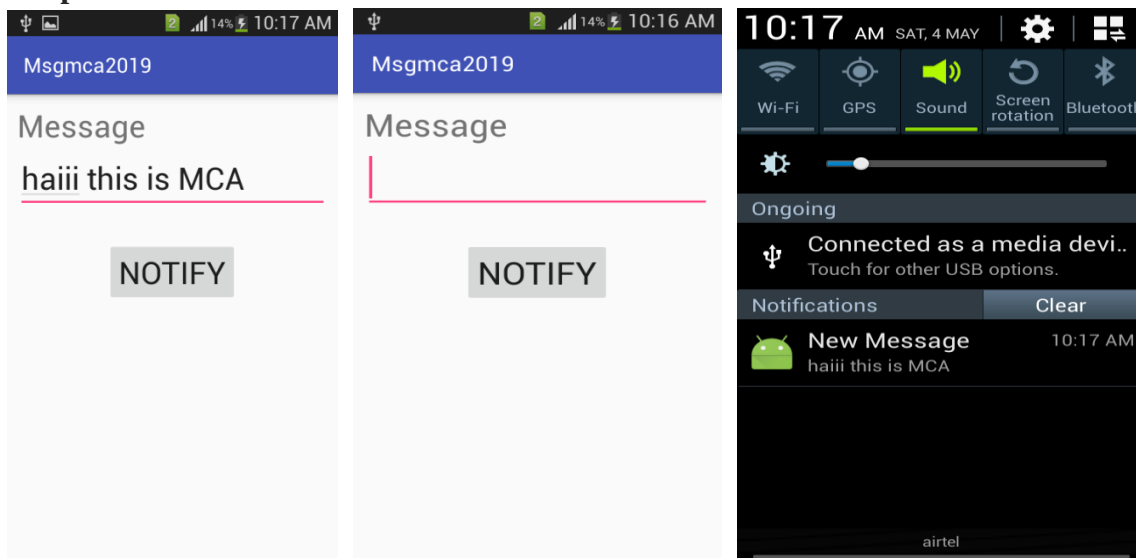
```

        notify= (Button) findViewById(R.id.button);
        e= (EditText) findViewById(R.id.editText);

        notify.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                Intent intent = new Intent(MainActivity.this,
                SecondActivity.class);
                PendingIntent pending =
                PendingIntent.getActivity(MainActivity.this, 0, intent, 0);
                Notification noti =
                new Notification.Builder(MainActivity.this).setContentTitle("New
                Message").setContentText(e.getText().toString()).setSmallIcon(R.mipmap.ic_launc
                her).setContentIntent(pending).build();
                NotificationManager manager = (NotificationManager)
                getSystemService(NOTIFICATION_SERVICE);
                noti.flags |= Notification.FLAG_AUTO_CANCEL;
                manager.notify(0, noti);
            }
        });
    }
}

```

Output :



Program : 7

Android Application that creates Alarm Clock

Aim:

To develop an Android Application that creates Alarm Clock.

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"/>

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:layout_margin="20dp"
        android:checked="false"
        android:onClick="OnToggleClicked"/>

</LinearLayout>
```

Changes in Manifest for the Android Application:

- Click on **app -> manifests -> AndroidManifest.xml**

Code for AndroidManifest.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<manifestxmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.mca.alarmclock">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:supportRtl="true"
        android:theme="@style/AppTheme">
        <activityandroid:name=".MainActivity">
            <intent-filter>
                <actionandroid:name="android.intent.action.MAIN"/>

                <categoryandroid:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
        <receiverandroid:name=".AlarmReceiver">
        </receiver>
    </application>

</manifest>
```

Code for MainActivity.java:

```
import android.app.AlarmManager;
```

```

import android.app.PendingIntent;
import android.content.Intent;
import android.os.Bundle;
import android.support.v7.app.AppCompatActivity;
import android.view.View;
import android.widget.TimePicker;
import android.widget.Toast;
import android.widget.ToggleButton;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity
{
    TimePicker alarmTimePicker;
    PendingIntent pendingIntent;
    AlarmManager alarmManager;

    @Override
    Protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        alarmTimePicker = (TimePicker) findViewById(R.id.timePicker);
        alarmManager = (AlarmManager) getSystemService(ALARM_SERVICE);
    }
    public void OnToggleClicked(View view)
    {
        longtime;
        if(((ToggleButton) view).isChecked())
        {
            Toast.makeText(MainActivity.this, "ALARM ON",
Toast.LENGTH_SHORT).show();
            Calendar calendar = Calendar.getInstance();
            calendar.set(Calendar.HOUR_OF_DAY,
alarmTimePicker.getCurrentHour());
            calendar.set(Calendar.MINUTE, alarmTimePicker.getCurrentMinute());
            Intent intent = new Intent(this, AlarmReceiver.class);
            pendingIntent = PendingIntent.getBroadcast(this, 0, intent, 0);

            time=(calendar.getTimeInMillis()-
(calendar.getTimeInMillis()%60000));
            if(System.currentTimeMillis()>time)
            {
                if(calendar.AM_PM == 0)
                    time = time + (1000*60*60*12);
                else
                    time = time + (1000*60*60*24);
            }
            alarmManager.setRepeating(AlarmManager.RTC_WAKEUP, time, 10000,
pendingIntent);
        }
        else
        {
            alarmManager.cancel(pendingIntent);
            Toast.makeText(MainActivity.this, "ALARM OFF",
Toast.LENGTH_SHORT).show();
        }
    }
}

```

Java Coding for Alarm Receiver:

- Click on app -> java -> com.example.mca.alarmclock -> AlarmReceiver.

Code for AlarmReceiver.java:

```

import android.content.BroadcastReceiver;
import android.content.Context;

```

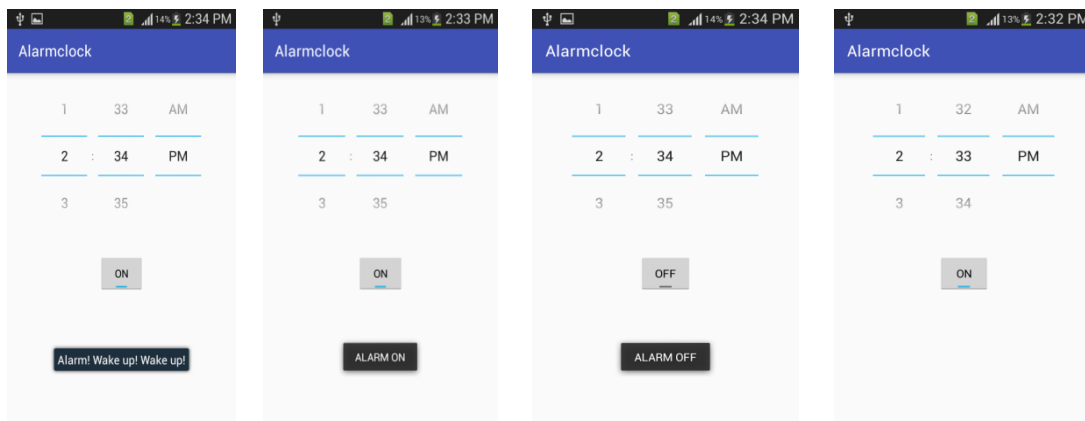
```

import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;
import android.widget.Toast;

public class AlarmReceiver extends BroadcastReceiver
{
    @Override
    public void onReceive(Context context, Intent intent)
    {
        Toast.makeText(context, "Alarm! Wake up! Wake up!",
        Toast.LENGTH_LONG).show();
        Uri alarmUri =
        RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
        if(alarmUri == null)
        {
            alarmUri =
            RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
        }
        Ringtone ringtone = RingtoneManager.getRingtone(context, alarmUri);
        ringtone.play();
    }
}

```

Output



Program : 8

Read/Write Data to internal storage

Aim:

To develop an Android Application reads/writes data from/to internal storage.

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    >

    <TextView

        android:id="@+id/textView1"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:layout_alignParentLeft="true"

        android:layout_alignParentRight="true"

        android:padding="5dp"

        android:text="Android Read and Write Text from/to a File"

        android:textStyle="bold"

        android:textSize="28sp" />

    <EditText

        android:id="@+id/editText1"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:layout_alignParentLeft="true"

        android:layout_alignParentRight="true"

        android:layout_below="@+id/textView1"

        android:layout_marginTop="22dp"

        android:minLines="5"

        android:layout_margin="5dp">
```



```

<requestFocus />

</EditText>

<Button
    android:id="@+id/button1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Write Text into File"
    android:onClick="WriteBtn"
    android:layout_alignTop="@+id/button2"
    android:layout_alignRight="@+id/editText1"
    android:layout_alignEnd="@+id/editText1" />

<Button
    android:id="@+id/button2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Read Text From file"
    android:onClick="ReadBtn"
    android:layout_centerVertical="true"
    android:layout_alignLeft="@+id/editText1"
    android:layout_alignStart="@+id/editText1" />

</RelativeLayout>

```

Code for MainActivity.java:

```

package com.example.mca.interst;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.InputStreamReader;
import java.io.OutputStreamWriter;

```

```

public class MainActivity extends Activity {

    EditText textmsg;

    static final int READ_BLOCK_SIZE = 100;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        textmsg=(EditText) findViewById(R.id.editText1);

    }

    // write text to file

    public void WriteBtn(View v) {

        // add-write text into file

        try {

            FileOutputStream fileout=openFileOutput("mytextfile.txt",
            MODE_PRIVATE);

            OutputStreamWriter outputWriter=new OutputStreamWriter(fileout);

            outputWriter.write(textmsg.getText().toString());

            outputWriter.close();

            //display file saved message

            Toast.makeText(getApplicationContext(), "File saved successfully!",

                Toast.LENGTH_SHORT).show();

        } catch (Exception e) {

            e.printStackTrace();

        }

    }

    // Read text from file

    public void ReadBtn(View v) {

        //reading text from file

        try {

            FileInputStream fileIn=openFileInput("mytextfile.txt");

            InputStreamReader InputRead= new InputStreamReader(fileIn);

```

```

char[] inputBuffer= new char[READ_BLOCK_SIZE];

String s="";

int charRead;

while ((charRead=InputRead.read(inputBuffer))>0) {

    // char to string conversion
    String readstring=String.valueOf(inputBuffer,0,charRead);

        s +=readstring;

    }

    InputRead.close();

    textmsg.setText(s);

} catch (Exception e) {

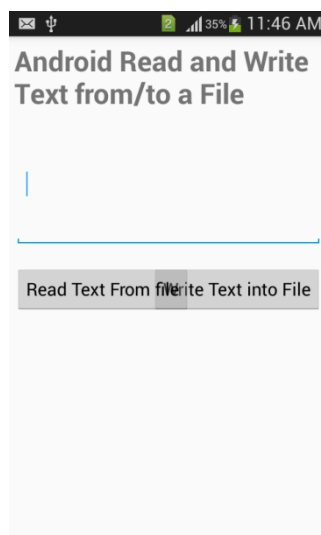
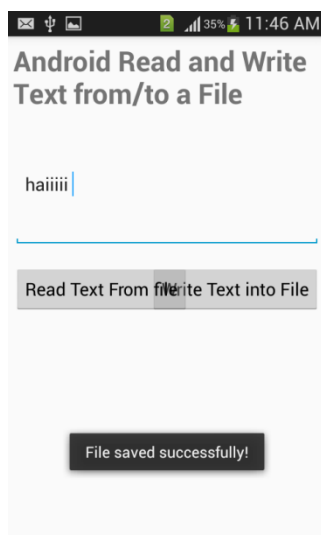
    e.printStackTrace();

}

}
}

```

Output :



Program : 9

Simple Android Application For Employee Details that makes use of Database

Aim:

To develop an Android Application that make use of database for employee details handling

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"

    android:layout_height="match_parent"
    android:paddingLeft="@dimen/activity_horizontal_margin"

    android:paddingRight="@dimen/activity_horizontal_margin"

    android:paddingTop="@dimen/activity_vertical_margin"

    android:paddingBottom="@dimen/activity_vertical_margin"
    tools:context=".MainActivity">

    <EditText

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:id="@+id/edtempleeenname"

        android:text="Search"

        android:layout_alignRight="@+id/btnupdate"

        android:layout_alignEnd="@+id/btnupdate"

        android:layout_alignParentLeft="true"

        android:layout_alignParentStart="true" />

    <EditText

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:id="@+id/editText"

        android:text="Employee Name"

        android:layout_below="@+id/edtempleeenname"

        android:layout_alignParentRight="true"

        android:layout_alignParentEnd="true"

        android:layout_alignParentLeft="true"

        android:layout_alignParentStart="true" />

    <EditText
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/editText2"
        android:layout_below="@+id/editText"
        android:layout_alignParentLeft="true"
        android:layout_alignParentStart="true"
        android:layout_alignRight="@+id/editText"
        android:layout_alignEnd="@+id/editText"
        android:text="Employee MailId" />
```

<EditText

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/editText3"
        android:layout_below="@+id/editText2"
        android:layout_alignParentLeft="true"
        android:layout_alignParentStart="true"
        android:layout_alignRight="@+id/editText2"
        android:layout_alignEnd="@+id/editText2"
        android:text="Employee Salary" />
```

<Button

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Save"
        android:id="@+id/btnsave"
        android:layout_centerVertical="true"
        android:layout_alignParentLeft="true"
        android:layout_alignParentStart="true" />
```

<Button

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Update"
        android:id="@+id/btnupdate"
        android:layout_alignTop="@+id/btnsave"
        android:layout_alignParentRight="true"
        android:layout_alignParentEnd="true" />
```

<Button

```
        android:layout_width="wrap_content"
```

```

        android:layout_height="wrap_content"

        android:text="Delete"

        android:id="@+id/btnndel"

        android:layout_centerVertical="true"

        android:layout_centerHorizontal="true" />
<Button

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Search All Record"

        android:id="@+id/btnselect"

        android:layout_alignTop="@+id/btnselectpeticular"

        android:layout_alignLeft="@+id/btnndel"

        android:layout_alignStart="@+id/btnndel" />

<Button

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Search"

        android:id="@+id/btnselectpeticular"

        android:layout_below="@+id/btnsave"

        android:layout_alignParentLeft="true"

        android:layout_alignParentStart="true"

        android:layout_marginTop="53dp" />

</RelativeLayout>

```

Code for MainActivity.java:

```

package com.example.mca.sqdb;

import android.content.Context;

import android.database.Cursor;

import android.database.sqlite.SQLiteDatabase;

import android.support.v7.app.AppCompatActivity;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.Toast;

public class MainActivity extends AppCompatActivity implements
android.view.View.OnClickListener {

```

```

        SQLiteDatabase db;

        EditText editsearchname,editempname,editempmail,editempsalary;

        Button Add, Delete, Modify, View,search ;

        @Override

        protected void onCreate(Bundle savedInstanceState) {

            super.onCreate(savedInstanceState);

            setContentView(R.layout.activity_main);

            //Create database,EmployeeDB database name

            db=openOrCreateDatabase("EmployeeDB", Context.MODE_PRIVATE, null);

            //create table Employee

            db.execSQL("CREATE TABLE IF NOT EXISTS Employee(EmpId INTEGER PRIMARY KEY
            AUTOINCREMENT,EmpName VARCHAR,EmpMail VARCHAR,EmpSalary VARCHAR)");

            editsearchname = (EditText) findViewById(R.id.edtemployeeaname);

            editempname = (EditText) findViewById(R.id.editText);

            editempmail = (EditText) findViewById(R.id.editText2);

            editempsalary = (EditText) findViewById(R.id.editText3);

            Add = (Button) findViewById(R.id.btnsave);

            Delete= (Button) findViewById(R.id.btndel);

            Modify= (Button) findViewById(R.id.btnupdate);

            View= (Button) findViewById(R.id. btnselect);

            search=(Button) findViewById(R.id. btnselectparticular);

            Add.setOnClickListener(this);

            Delete.setOnClickListener(this);

            Modify.setOnClickListener(this);

            View.setOnClickListener(this);

            search.setOnClickListener(this);

        }

        public void msg(Context context,String str)

        {

            Toast.makeText(this,str,Toast.LENGTH_SHORT).show();

        }

        @Override

        public void onClick(View v) {

            if(v.getId()==R.id.btnsave)

            {

```

```

// code for save data

if(editempname.getText().toString().trim().length()==0||

        editempmail.getText().toString().trim().length()==0||

        editempsalary.getText().toString().trim().length()==0)

    {

        msg(this, "Please enter all values");

return;

    }

    db.execSQL("INSERT INTO Employee(EmpName,EmpMail,EmpSalary)VALUES('"+
editempname.getText()+"','"+ editempmail.getText()+ "','"+
editempsalary.getText()+"')");

        msg(this, "Record added");

    }


else if(v.getId()==R.id.btnupdate)

    {

//code for update data

if(editsearchname.getText().toString().trim().length()==0)

    {

        msg(this, "Enter Employee Name");

return;

    }

        Cursor c=db.rawQuery("SELECT * FROM Employee WHERE EmpName='"+
editsearchname.getText()+"'", null);

if(c.moveToFirst()) {

        db.execSQL("UPDATE Employee SET EmpName='"+
editempname.getText()+"', EmpMail='"+ editempmail.getText()+"',EmpSalary='"+
editempsalary.getText()+"' WHERE EmpName='"+editsearchname.getText()+"'");

        msg(this, "Record Modified");

    }

else

{

        msg(this, "Invalid Employee Name");

    }

}

else if(v.getId()==R.id.btndel)

    {

//code for delete data

if(editsearchname.getText().toString().trim().length()==0)

    {

```



```

        msg(this, " Please enter Employee Name ");

return;

    }

    Cursor c=db.rawQuery("SELECT * FROM Employee WHERE EmpName ='"+
editsearchname.getText()+"'", null);

    if(c.moveToFirst())

    {

        db.execSQL("DELETE FROM Employee WHERE EmpName ='"+
editsearchname.getText()+"'");

        msg(this, "Record Deleted");

    }

else

{

        msg(this, "Invalid Employee Name ");

    }

}

else if (v.getId() == R.id.btnselect)

{

    //code for select all data

    Cursor c=db.rawQuery("SELECT * FROM Employee", null);

    if(c.getCount()==0)

    {

        msg(this, "No records found");

return;

    }

    StringBuffer buffer=new StringBuffer();

while(c.moveToNext())

    {

        buffer.append("Employee Name: "+c.getString(1)+"\n");

        buffer.append("Employee Mail: "+c.getString(2)+"\n\n");

        buffer.append("Employee Salary: "+c.getString(3)+"\n\n");

    }

    msg(this, buffer.toString());

}

else if(v.getId()==R.id.btnselectparticular)

{

    //code for select particular data

    if(editsearchname.getText().toString().trim().length()==0)

```

```

    {
        msg(this, "Enter Employee Name");

return;
    }

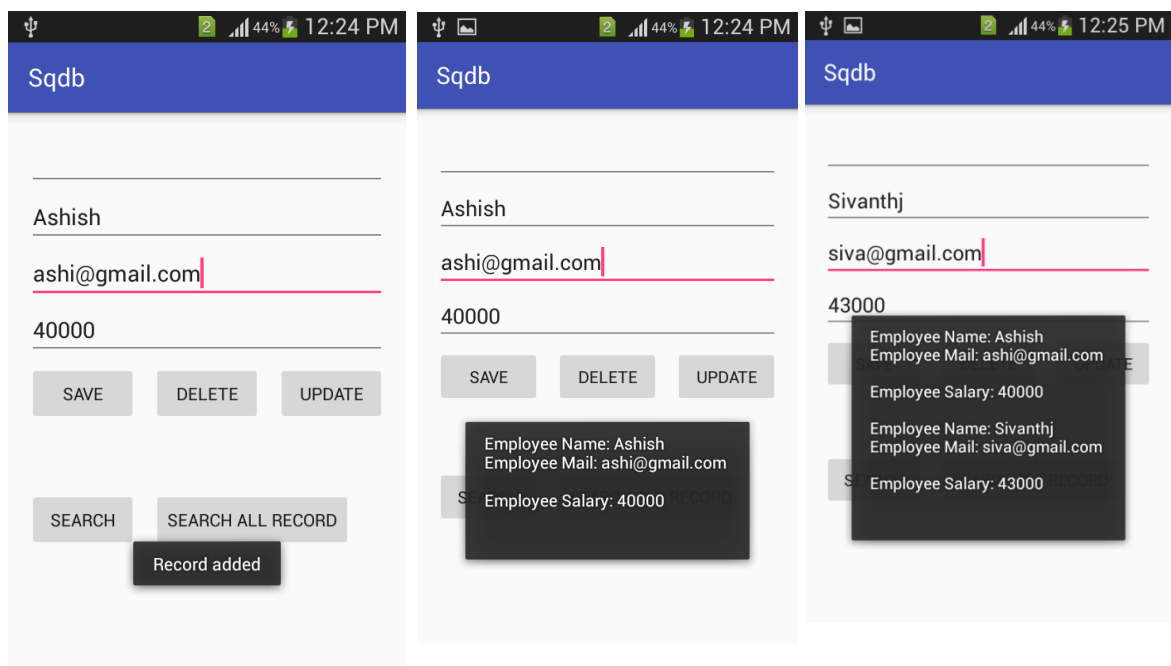
    Cursor c=db.rawQuery("SELECT * FROM Employee WHERE
EmpName='"+editsearchname.getText()+"'", null);

    if(c.moveToFirst())
    {
        editempname.setText(c.getString(1));
        editempmail.setText(c.getString(2));
        editempsalary.setText(c.getString(3));
    }

else
{
    msg(this, "Invalid Employee Name");
}
}
}
}

```

Output :



Program : 10

Simple Android Application For Progress Bar

Aim:

To develop an Android Application that show the working of progress bar

Code for Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".MainActivity" >

    <ProgressBar

        android:id="@+id/progressBar"

        style="?android:attr/progressBarStyleHorizontal"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:layout_alignParentLeft="true"

        android:layout_alignParentTop="true"

        android:layout_marginLeft="23dp"

        android:layout_marginTop="20dp"

        android:indeterminate="false"

        android:max="100"

        android:minHeight="50dp"

        android:minWidth="200dp"

        android:progress="1" />

    <ProgressBar

        android:id="@+id/progressBar_cyclic"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:minHeight="50dp"

        android:minWidth="50dp"

        android:layout_centerVertical="true"
```

```

        android:layout_centerHorizontal="true" />

<TextView
    android:id="@+id/textView"

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:layout_alignLeft="@+id/progressBar"

    android:layout_below="@+id/progressBar"/>

</RelativeLayout>

```

Code for MainActivity.java:

```

package com.example.mca.progress;

import android.os.Handler;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ProgressBar;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    private ProgressBar progressBar;
    private int progressStatus = 0;
    private TextView textView;
    private Handler handler = new Handler();

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        progressBar = (ProgressBar) findViewById(R.id.progressBar);

        textView = (TextView) findViewById(R.id.textView);

        // Start long running operation in a background thread
        new Thread(new Runnable() {

            public void run() {

                while (progressStatus < 100) {

                    progressStatus += 1;

```

```

// Update the progress bar and display the
//current value in the text view

handler.post(new Runnable() {

    public void run() {

        progressBar.setProgress(progressStatus);

        textView.setText(progressStatus+"/"+progressBar.getMax());

    }

});

try {
    // Sleep for 200 milliseconds.
    Thread.sleep(200);

    } catch (InterruptedException e) {
        e.printStackTrace();
    }

}

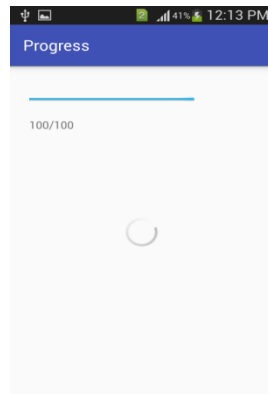
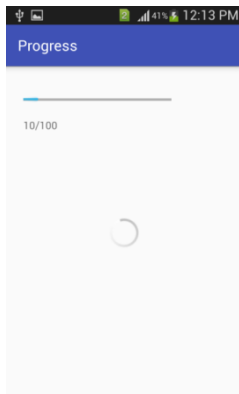
}).start();

}

}

```

Output :



Program : 11

Simple Android Application for Showing Animation

Aim:

To develop an Android Application For Showing Animation

Code for Activity_main.xml:

```
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent" >
```

```
<RelativeLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
<Button
    android:id="@+id/btnFadeIn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:text="Fade In" />
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textAppearance="?android:attr/textAppearanceMedium"
    android:text="Fade In"
    android:id="@+id/txt_fade_in"
    android:layout_alignBottom="@+id/btnFadeIn"
    android:layout_alignLeft="@+id/txt_fade_out"
    android:layout_alignStart="@+id/txt_fade_out" />
```

```
<Button
    android:id="@+id/btnFadeOut"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:layout_below="@id/btnFadeIn"
    android:text="Fade Out" />
```

```
<Button
    android:id="@+id/btnCrossFade"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:layout_below="@id/btnFadeOut"
    android:text="Cross Fade" />
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

```
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Cross Fade In"
    android:id="@+id/txt_out"
android:visibility="gone"
android:layout_gravity="center_horizontal"
    android:layout_alignTop="@+id/txt_in"
    android:layout_alignLeft="@+id/txt_in"
    android:layout_alignStart="@+id/txt_in" />
```

```
<Button
    android:id="@+id/btnBlink"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:layout_below="@id/btnCrossFade"
    android:text="Blink" />
```

```
<Button
    android:id="@+id/btnZoomIn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:layout_below="@id/btnBlink"
    android:text="Zoom In" />
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textAppearance="?android:attr/textAppearanceMedium"
    android:text="Blink"
    android:id="@+id/txt_blink"
    android:layout_gravity="center_horizontal"
    android:layout_alignBottom="@+id/btnBlink"
    android:layout_alignLeft="@+id/txt_zoom_in"
    android:layout_alignStart="@+id/txt_zoom_in" />
```

```
<Button
    android:id="@+id/btnZoomOut"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:layout_below="@id/btnZoomIn"
    android:text="Zoom Out" />
```

```
<Button
    android:id="@+id/btnRotate"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:layout_below="@id/btnZoomOut"
    android:text="Rotate" />
```

```
<Button
    android:id="@+id/btnMove"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
```

```
        android:layout_below="@id/btnRotate"  
        android:text="Move" />
```

```
<Button  
    android:id="@+id/btnSlideUp"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_margin="5dp"  
    android:layout_below="@id/btnMove"  
    android:text="Slide Up" />
```

```
<TextView  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:textAppearance="?android:attr/textAppearanceMedium"  
    android:text="Fade Out"  
    android:id="@+id/txt_fade_out"  
    android:layout_gravity="center_horizontal"  
    android:layout_alignBottom="@+id/btnFadeOut"  
    android:layout_alignLeft="@+id/txt_in"  
    android:layout_alignStart="@+id/txt_in" />
```

```
<Button  
    android:id="@+id/btnSlideDown"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_margin="5dp"  
    android:layout_below="@id/btnSlideUp"  
    android:text="Slide Down" />
```

```
<Button  
    android:id="@+id/btnBounce"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_margin="5dp"  
    android:layout_below="@id/btnSlideDown"  
    android:text="Bounce" />
```

```
<Button  
    android:id="@+id/btnSequential"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_margin="5dp"  
    android:layout_below="@id/btnBounce"  
    android:text="Sequential Animation" />
```

```
<Button  
    android:id="@+id/btnTogether"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_below="@id/btnSequential"  
    android:layout_margin="5dp"  
    android:text="Together Animation" />
```

```
<TextView  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"
```



```
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Cross Fade Out"
    android:id="@+id/txt_in"
android:layout_gravity="center_horizontal"
    android:layout_alignBottom="@+id/btnCrossFade"
    android:layout_alignLeft="@+id/txt_blink"
    android:layout_alignStart="@+id/txt_blink" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Zoom In"
    android:id="@+id/txt_zoom_in"
    android:layout_alignBottom="@+id/btnZoomIn"
    android:layout_alignLeft="@+id/txt_zoom_out"
    android:layout_alignStart="@+id/txt_zoom_out" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Zoom Out"
    android:id="@+id/txt_zoom_out"
    android:layout_alignBottom="@+id/btnZoomOut"
    android:layout_toRightOf="@+id/btnSequential"
    android:layout_toEndOf="@+id/btnSequential" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Rotate"
    android:id="@+id/txt_rotate"
    android:layout_above="@+id/btnMove"
    android:layout_toRightOf="@+id/btnSequential"
    android:layout_toEndOf="@+id/btnSequential" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Move"
    android:id="@+id/txt_move"
    android:layout_alignBottom="@+id/btnMove"
    android:layout_alignLeft="@+id/txt_slide_up"
    android:layout_alignStart="@+id/txt_slide_up" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Slide Up"
    android:id="@+id/txt_slide_up"
    android:layout_alignBottom="@+id/btnSlideUp"
    android:layout_toRightOf="@+id/btnSequential"
    android:layout_toEndOf="@+id/btnSequential" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Slide Down"
    android:id="@+id/txt_slide_down"
    android:layout_alignBottom="@+id/btnSlideDown"
    android:layout_alignLeft="@+id/txt_slide_up"
    android:layout_alignStart="@+id/txt_slide_up" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Bounce"
    android:id="@+id/txt_bounce"
    android:layout_alignBottom="@+id/btnBounce"
    android:layout_alignLeft="@+id/txt_slide_down"
    android:layout_alignStart="@+id/txt_slide_down" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Sequential"
    android:id="@+id/txt_seq"
    android:layout_alignBottom="@+id/btnSequential"
    android:layout_alignLeft="@+id/txt_bounce"
    android:layout_alignStart="@+id/txt_bounce" />
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textAppearance="?android:attr/textAppearanceMedium"
android:text="Together"
    android:id="@+id/txt_tog"
    android:layout_alignBottom="@+id/btnTogether"
    android:layout_toRightOf="@+id/btnSequential"
    android:layout_toEndOf="@+id/btnSequential" />
```

```
</RelativeLayout>
```

```
</ScrollView>
```

fade_in.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
android:fillAfter="true" >
```

```
<alpha
android:duration="1000"
android:fromAlpha="0.0"
    android:interpolator="@android:anim/accelerate_interpolator"
android:toAlpha="1.0" />
```

```
</set>
```

fade_out.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:fillAfter="true" >
```

```
<alpha
    android:duration="1000"
    android:fromAlpha="1.0"
        android:interpolator="@android:anim/accelerate_interpolator"
    android:toAlpha="0.0" />
```

```
</set>
```

blink.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android">
<alpha android:fromAlpha="0.0"
    android:toAlpha="1.0"
        android:interpolator="@android:anim/accelerate_interpolator"
    android:duration="600"
    android:repeatMode="reverse"
    android:repeatCount="infinite"/>
</set>
```

zoom_in.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:fillAfter="true" >
```

```
<scale
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:duration="1000"
    android:fromXScale="1"
    android:fromYScale="1"
    android:pivotX="50%"
    android:pivotY="50%"
    android:toXScale="3"
    android:toYScale="3" >
</scale>
```

```
</set>
```

zoom_out.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:fillAfter="true" >
```

```
<scale
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:duration="1000"
    android:fromXScale="1.0"
    android:fromYScale="1.0"
    android:pivotX="50%"
    android:pivotY="50%"
    android:toXScale="0.5"
```

```
android:toYScale="0.5" >
</scale>
```

```
</set>
```

rotate.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android">
<rotate android:fromDegrees="0"
android:toDegrees="360"
android:pivotX="50%"
android:pivotY="50%"
android:duration="600"
android:repeatMode="restart"
android:repeatCount="infinite"
    android:interpolator="@android:anim/cycle_interpolator"/>

</set>
```

move.xml

```
<set
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:interpolator="@android:anim/linear_interpolator"
    android:fillAfter="true">

<translate
    android:fromXDelta="0%p"
    android:toXDelta="75%p"
    android:duration="800" />

</set>
```

slide_up.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:fillAfter="true" >

<scale
    android:duration="500"
    android:fromXScale="1.0"
    android:fromYScale="1.0"
        android:interpolator="@android:anim/linear_interpolator"
    android:toXScale="1.0"
    android:toYScale="0.0" />

</set>
```

slide_down.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:fillAfter="true">

<scale
    android:duration="500"
    android:fromXScale="1.0"
    android:fromYScale="0.0"
    android:toXScale="1.0"
    android:toYScale="1.0" />
```

```
</set>
```

bounce.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:fillAfter="true"
        android:interpolator="@android:anim/bounce_interpolator">

<scale
    android:duration="500"
    android:fromXScale="1.0"
    android:fromYScale="0.0"
    android:toXScale="1.0"
    android:toYScale="1.0" />

</set>
```

sequential.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:fillAfter="true"
        android:interpolator="@android:anim/linear_interpolator" >

<!-- Move -->
<translate
    android:duration="800"
    android:fillAfter="true"
    android:fromXDelta="0%p"
    android:startOffset="300"
    android:toXDelta="75%p" />
<translate
    android:duration="800"
    android:fillAfter="true"
    android:fromYDelta="0%p"
    android:startOffset="1100"
    android:toYDelta="70%p" />
<translate
    android:duration="800"
    android:fillAfter="true"
    android:fromXDelta="0%p"
    android:startOffset="1900"
    android:toXDelta="-75%p" />
<translate
    android:duration="800"
    android:fillAfter="true"
    android:fromYDelta="0%p"
    android:startOffset="2700"
    android:toYDelta="-70%p" />

<!-- Rotate 360 degrees -->
<rotate
    android:duration="1000"
    android:fromDegrees="0"
        android:interpolator="@android:anim/cycle_interpolator"
    android:pivotX="50%"
    android:pivotY="50%"
    android:startOffset="3800"
    android:repeatCount="infinite"
    android:repeatMode="restart"
    android:toDegrees="360" />

</set>
```

together.xml

```
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:fillAfter="true"
    android:interpolator="@android:anim/linear_interpolator" >

<!-- Move -->
<scale
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:duration="4000"
    android:fromXScale="1"
    android:fromYScale="1"
    android:pivotX="50%"
    android:pivotY="50%"
    android:toXScale="4"
    android:toYScale="4" >

<!-- Rotate 180 degrees -->
<rotate
    android:duration="500"
    android:fromDegrees="0"
    android:pivotX="50%"
    android:pivotY="50%"
    android:repeatCount="infinite"
    android:repeatMode="restart"
    android:toDegrees="360" />

</set>
```

Code for MainActivity.java:

```
import android.app.Activity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends Activity {

    Button btnFadeIn, btnFadeOut, btnCrossFade, btnBlink, btnZoomIn,
    btnZoomOut, btnRotate, btnMove, btnSlideUp, btnSlideDown,
    btnBounce, btnSequential, btnTogether;

    Animation
    animFadeIn, animFadeOut, animBlink, animZoomIn, animZoomOut, animRotate
    , animMove, animSlideUp, animSlideDown, animBounce, animSequential, animTogether,
    animCrossFadeIn, animCrossFadeOut;

    TextView
    txtFadeIn, txtFadeOut, txtBlink, txtZoomIn, txtZoomOut, txtRotate, txtMove, txtSlideUp,
    txtSlideDown, txtBounce, txtSeq, txtTog, txtIn, txtOut;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnFadeIn = (Button) findViewById(R.id.btnFadeIn);
        btnFadeOut = (Button) findViewById(R.id.btnFadeOut);
```

```

btnCrossFade = (Button) findViewById(R.id.btnCrossFade);
btnBlink = (Button) findViewById(R.id.btnBlink);
btnZoomIn = (Button) findViewById(R.id.btnZoomIn);
btnZoomOut = (Button) findViewById(R.id.btnZoomOut);
btnRotate = (Button) findViewById(R.id.btnRotate);
btnMove = (Button) findViewById(R.id.btnMove);
btnSlideUp = (Button) findViewById(R.id.btnSlideUp);
btnSlideDown = (Button) findViewById(R.id.btnSlideDown);
btnBounce = (Button) findViewById(R.id.btnBounce);
btnSequential = (Button) findViewById(R.id.btnSequential);
btnTogether = (Button) findViewById(R.id.btnTogether);
txtFadeIn=(TextView)findViewById(R.id.txt_fade_in);
txtFadeOut=(TextView)findViewById(R.id.txt_fade_out);
txtBlink=(TextView)findViewById(R.id.txt_blink);
txtZoomIn=(TextView)findViewById(R.id.txt_zoom_in);
txtZoomOut=(TextView)findViewById(R.id.txt_zoom_out);
txtRotate=(TextView)findViewById(R.id.txt_rotate);
txtMove=(TextView)findViewById(R.id.txt_move);
txtSlideUp=(TextView)findViewById(R.id.txt_slide_up);
txtSlideDown=(TextView)findViewById(R.id.txt_slide_down);
txtBounce=(TextView)findViewById(R.id.txt_bounce);
txtSeq=(TextView)findViewById(R.id.txt_seq);
txtTog=(TextView)findViewById(R.id.txt_tog);
txtIn=(TextView)findViewById(R.id.txt_in);
txtOut=(TextView)findViewById(R.id.txt_out);
animFadeIn = AnimationUtils.loadAnimation(getApplicationContext(),
    R.anim.fade_in);

animFadeIn = AnimationUtils.loadAnimation(getApplicationContext(),
    R.anim.fade_in);

// fade in
btnFadeIn.setOnClickListener(new View.OnClickListener() {

@Override
public void onClick(View v){
txtFadeIn.setVisibility(View.VISIBLE);
txtFadeIn.startAnimation(animFadeIn);
    }

});

animFadeOut = AnimationUtils.loadAnimation(getApplicationContext(),
    R.anim.fade_out);

// fade out
btnFadeOut.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v){
txtFadeOut.setVisibility(View.VISIBLE);
txtFadeOut.startAnimation(animFadeOut);
    }

});

animCrossFadeIn = AnimationUtils.loadAnimation(getApplicationContext(),
    R.anim.fade_in);
animCrossFadeOut = AnimationUtils.loadAnimation(getApplicationContext(),
    R.anim.fade_out);

// cross fade
btnCrossFade.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v){
txtOut.setVisibility(View.VISIBLE);
// start fade in animation
txtOut.startAnimation(animCrossFadeIn);

// start fade out animation
txtIn.startAnimation(animCrossFadeOut);
    }

});

animBlink = AnimationUtils.loadAnimation(getApplicationContext(),

```

```

        R.anim.blink);

// blink
btnBlink.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
txtBlink.setVisibility(View.VISIBLE);
txtBlink.startAnimation(animBlink);
        }
});

animZoomIn = AnimationUtils.loadAnimation(getApplicationContext(),
        R.anim.zoom_in);

// Zoom In
btnZoomIn.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
txtZoomIn.setVisibility(View.VISIBLE);
txtZoomIn.startAnimation(animZoomIn);
        }
});

animZoomOut = AnimationUtils.loadAnimation(getApplicationContext(),
        R.anim.zoom_out);

// Zoom Out
btnZoomOut.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
txtZoomOut.setVisibility(View.VISIBLE);
txtZoomOut.startAnimation(animZoomOut);
        }
});

animRotate = AnimationUtils.loadAnimation(getApplicationContext(),
        R.anim.rotate);

// Rotate
btnRotate.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
txtRotate.startAnimation(animRotate);
        }
});

animMove = AnimationUtils.loadAnimation(getApplicationContext(),
        R.anim.move);

// Move
btnMove.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
txtMove.startAnimation(animMove);
        }
});

animSlideUp = AnimationUtils.loadAnimation(getApplicationContext(),
        R.anim.slide_up);

// Slide Up
btnSlideUp.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
txtSlideUp.startAnimation(animSlideUp);
        }
});

animSlideDown = AnimationUtils.loadAnimation(getApplicationContext(),
        R.anim.slide_down);

// Slide Down
btnSlideDown.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
txtSlideDown.startAnimation(animSlideDown);
        }
});

animBounce = AnimationUtils.loadAnimation(getApplicationContext(),

```



```

        R.anim.bounce);
// Slide Down
btnBounce.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v){
txtBounce.startAnimation(animBounce);
        }
    });
animSequential = AnimationUtils.loadAnimation(getApplicationContext(),
        R.anim.sequential);
// Sequential
btnSequential.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v){

txtSeq.startAnimation(animSequential);
        }
    });
animTogether = AnimationUtils.loadAnimation(getApplicationContext(),
        R.anim.together);

// Together
btnTogether.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v){
txtTog.startAnimation(animTogether);
        }
    });
}
}

```

Output :

